MANIPAL INSTITUTE OF TECHNOLOGY BENGALURU

CSE 4191 – Minor Specialization Mini Project | Phase 1 Review



MANIPAL INSTITUTE OF TECHNOLOGY
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CLARA

Cloud-Linked Evaluation and Risk Assessment using Predictive Analytics and Business Intelligence

Johann Mathew Sajan, Nandan Murali Nair, Abhijay Gupta, Arshia Gali
Department of Computer Science and Engineering
Under the Guidance of: Dr. Geetabai S Hukkeri, Assistant Professor

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Abstract

CLARA (Cloud-Linked Evaluation and Risk Assessment) is a GenAI-powered cloud security and cost intelligence platform. Rapid cloud adoption has introduced unmanaged security misconfigurations and escalating cost inefficiencies — challenges that existing tools address in silos.

CLARA bridges this gap by integrating:

- A multi-dimensional cloud analytics **data warehouse** storing instance metadata, security configs, utilization metrics and cost data
- Three specialized **ML models** (Isolation Forest anomaly detection, Random Forest security scoring, regression-based cost prediction)
- A **Gemini API-powered NLG (Natural Language Generation) layer** translating risk scores into actionable recommendations
- An interactive **Streamlit BI dashboard** for risk heatmaps, cost trends, and recommendation management

The Problem: Cloud Governance Gap

28%
Avg cloud spend wasted (Flexera 2024)

19%
Cloud breaches from misconfigs (IBM 2023)

$1T+
Global cloud spend by 2026 (Gartner)

300+
Instances analyzed by CLARA

Security Challenges

- Cloud misconfigs are #1 cause of data breaches
- Open ports, missing encryption, over-permissioned IAM roles
- Misconfigs develop progressively over weeks — hard to detect
- No unified real-time visibility into security posture
- Rule-based scanners miss evolving threat patterns

Cost Challenges

- 28% average cloud spend wasted on idle resources
- No intelligent rightsizing recommendations
- Finance teams lack real-time analytics dashboards
- Underutilized instances go unidentified for months
- Existing tools don't bridge security AND cost simultaneously

Meet CLARA

A unified cloud intelligence platform combining security, cost, and AI-driven insights

DW

Data Warehouse

- PostgreSQL schema
- 7 normalized tables
- ETL pipeline
- 300+ instances, 90 days

ML

ML Risk Scoring

- Isolation Forest
- Random Forest (SHAP)
- Cost regression
- Score: 0–100

AI

GenAI Engine

- Gemini API
- Structured JSON output
- Priority: Crit/High/Med/Low
- NL explanations

BI

BI Dashboard

- 4-page Streamlit app
- Plotly visualizations
- Filters + export
- Recommendations UI

Literature Review

Cloud Security

Sharma et al., IEEE TCC 2022

Graph-based IAM misconfiguration detection — 40% faster than rule-based scanners

Configuration Drift

Patel & Kumar, ACM SoCC 2023

Temporal analysis shows misconfigs develop progressively; time-series monitoring is critical

Anomaly Detection

Zhang et al., IEEE BigData 2021

Isolation Forest achieves F1=0.89 on cloud workload telemetry; computationally efficient at scale

Ensemble Models

Liu et al., JNCA 2023

Isolation Forest + autoencoder ensemble improves anomaly detection by 12% over single models

Cost Optimization

Kim et al., IEEE Cloud 2023

Clustering-based approach identifies underutilized instance groups for automated termination

GenAI for IT Ops

Wang et al., ICSE 2024

LLM-powered recommendations reduce incident MTTR by 31% through contextual synthesis

Research Gap: No existing work unifies multi-model ML risk scoring, LLM recommendation generation, and interactive BI in a single cloud governance platform — CLARA addresses this gap.

Project Objectives

O1 Design a normalized 7-table cloud analytics data warehouse consolidating instance metadata, security configs, utilization metrics, cost records, and security events.

O2 Develop and validate 3 ML models: Isolation Forest (anomaly), Random Forest (security), and regression (cost) — generating risk scores 0–100 with >85% recall.

O3 Integrate Gemini API as an NLG layer converting risk scores into prioritized JSON recommendations with implementation steps and estimated cost savings.

O4 Build an interactive 4-page Streamlit BI dashboard with risk heatmaps, cost trends, vulnerability breakdowns, and a recommendation management interface.

O5 Generate a synthetic cloud dataset: 300+ instances, 90 days of metrics, intentional security vulnerabilities, and realistic cost patterns across AWS, Azure, and GCP.

O6 Develop automated test suite (>90% coverage) and prepare a Scopus/WoS-targeted research publication on cloud risk analytics and AI-driven governance.

System Architecture

Layer 1

Data Layer

- Synthetic data gen
- PostgreSQL warehouse
- 7 normalized tables
- ETL pipeline (Python)

Layer 2

Analytics Layer

- Isolation Forest (Anomaly)
- Random Forest (Security)
- Regression (Cost)
- SHAP interpretability

Layer 3

GenAI Layer

- Gemini API
- Recommendation JSON
- Priority scoring
- Insights engine

Layer 4

Presentation Layer

- Streamlit (4 pages)
- Plotly charts
- Filters + export
- Audit trail

▶ *Data Flow: Synthetic Data → Warehouse → ML Risk Scores → GenAI Recommendations → Dashboard*

ML Models & Dataset Design

Machine Learning Models

Model	Algorithm	Output
Anomaly Detector	Isolation Forest	anomaly_score 0-100
Security Scorer	Random Forest + SHAP	security_risk_score 0-100
Cost Predictor	Linear Regression + K-Means	waste_score + savings $
Overall Score	Weighted Average	overall_risk_score 0-100

Features Used

- Security: Open ports, missing encryption, IAM permissions, failed logins
- Anomaly: CPU/memory variance, network spikes, event frequency
- Cost: Utilization %, instance age, hourly rate, last access date

Synthetic Dataset

300+
Cloud Instances

90
Days of Metrics

5
Data Tables

7
Warehouse Tables

Intentional Vulnerabilities Injected:

- 10% instances: SSH exposed to 0.0.0.0/0
- 5% instances: No storage encryption
- 15% instances: Excessive IAM permissions
- 20% instances: Consistently underutilized (<5%)

Technology Stack & Team Roles

Technology Stack

Language	Python 3.10+
Data Warehouse	PostgreSQL / DuckDB
ML / Analytics	scikit-learn, River, SHAP
GenAI	Google Gemini API
Dashboard	Streamlit, Plotly
ETL	Pandas, SQLAlchemy
Testing / CI	Pytest, GitHub Actions
Version Control	Git, GitHub

Team Contributions

Abhijay Gupta

Data Warehouse & Analytics Lead

DB schema, ETL pipeline, ML models

Nandan Murali Nair

BI Dashboard Lead

Streamlit app, Plotly charts, UX

Arshia Gali

GenAI Integration Lead

Gemini API, recommendations, insights

Johann Mathew Sajan

Synthetic Data & Testing Lead

Data generator, test suite, CI/CD

Conclusion & Roadmap

Phase 1 Summary

- Problem Statement & Motivation defined
- Thorough literature review of 8+ Scopus/WoS papers
- 7 specific, measurable project objectives set
- Complete 4-layer system architecture designed
- Normalized 7-table data warehouse schema finalized
- ML model selection and feature design complete
- Technology stack and team roles defined

Upcoming Phases

Phase 2

System architecture finalization, synopsis submission, initial ETL + DB setup

Phase 3

Full ML implementation, Gemini API integration, Streamlit dashboard, testing

Phase 4

Final demo, viva, research paper submission to Scopus/WoS conference

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CLARA

Thank You

Questions & Discussion

Project: **CLARA – Cloud-Linked Evaluation and Risk Assessment using Predictive Analytics and Business Intelligence**

Team: Johann Mathew Sajan, Nandan Murali Nair, Abhijay Gupta, Arshia Gali

Under the Guidance of: Dr. Geetabai S Hukkeri, Assistant Professor | Dept. of CSE, MIT Bengaluru | 2026–2027